

Change in Internal Energy During an Adiabatic Transformation

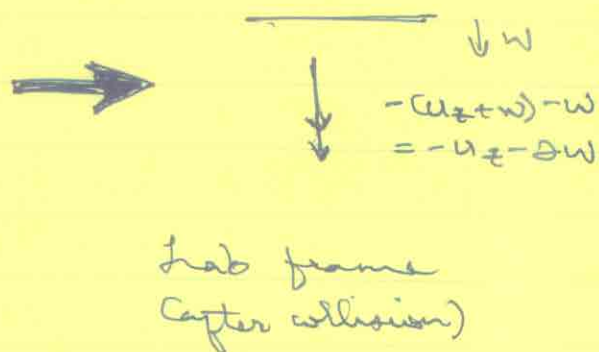
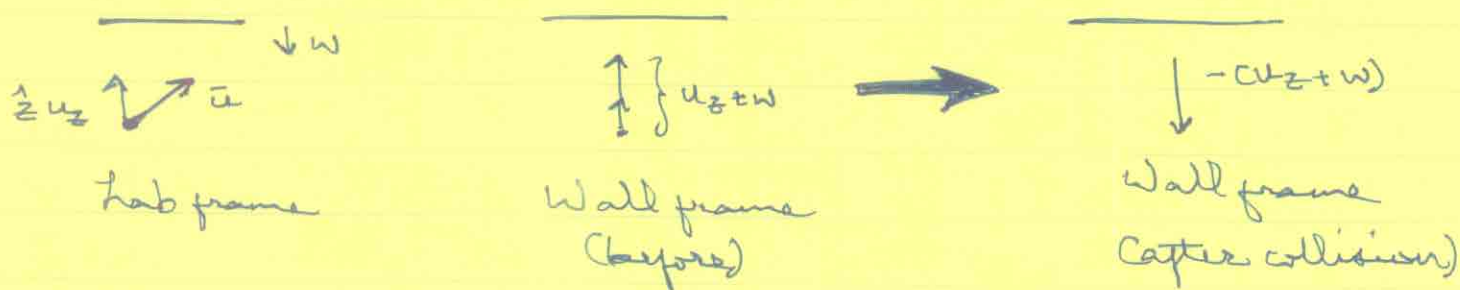


The vessel is filled with an ideal gas with some temperature and pressure.

Imagine a compression of the gas such that

$$\dot{h} = -w$$

Consider a gas particle with velocity $\vec{u}$ in the lab frame. Suppose that $u_z^{tw} > 0$, so that the particle is approaching the wall.



After the collision, the particle rebounds with velocity

$$\vec{u}' = (u_x, u_y, -u_z - 2w)$$

The new specific energy is

$$\begin{aligned}\frac{1}{2}u'^2 &= \frac{1}{2}(u_x^2 + u_y^2 + u_z^2 + 4u_zw + 4w^2) \\ &= \frac{1}{2}u^2 + 2w(u_z + w)\end{aligned}$$

$$\Rightarrow \text{excess energy} = 2w(\underline{u_z + w}) = \underline{\underline{\Delta E/m}}$$

Assume that the ~~change~~ change in h is sufficiently slow that the gas always remains in thermal equilibrium.

$$E = \text{total energy} = \frac{3}{2}NkT = \sum_{\text{all particles}} \frac{1}{2}mu^2$$

In a small time $\delta t \ll |h/\dot{h}|$:

$$\frac{\delta E}{A} = \delta t \int d^2u \int_{\substack{u_z + w > 0 \\ \text{net movement} \\ \text{toward wall}}} du_z f(u) n (\bar{u} + w\hat{z}) \cdot \hat{z} \Delta E$$

$$f(u) = \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-m\mathbf{v}^2/2kT}$$

$$\Rightarrow \frac{\delta E}{A} = \delta t n \int d^2u \int_{\substack{u_z + w > 0 \\ \text{assumed} \\ \text{constant} \\ \text{during} \\ \text{interval}}} du_z \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-m\mathbf{v}^2/2kT} (u_z + w) \times 2m(u_z w + w^2)$$

We can imagine the changes being arbitrarily slow, so that

$$(u_z + w) \cdot 2mw(u_z + w) \approx 2mw u_z^2 + O(w^2)$$

$$\begin{aligned} \Rightarrow \frac{\delta E}{A \delta t} &\approx \boxed{} n \cdot 2mw \int du_x du_y \int_0^\infty du_z \left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} e^{-\frac{mv^2}{2kT}} u_z^2 \\ &= 2m n w \left(\int_{-\infty}^\infty du e^{-\frac{mu^2}{2kT}} \right)^2 \left(\int_0^\infty du u^2 e^{-\frac{mu^2}{2kT}} \right) \left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} \\ &= 2m n w \left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} \left(\frac{2kT}{m}\right)^{\frac{3}{2}} \left(\frac{2kT}{m}\right) \\ &\quad \times \underbrace{\left(\int_{-\infty}^\infty du e^{-u^2} \right)^2}_{\sqrt{\pi} \cdot \sqrt{\pi} = \pi} \underbrace{\left(\int_0^\infty du u^2 e^{-u^2} \right)}_{\sqrt{\pi}/4} \\ &= \cancel{2m} n w \frac{2kT}{m} \cdot \frac{1}{4} = n w \cdot kT \end{aligned}$$

$$\begin{aligned} \Rightarrow \frac{dE}{dt} &= \underbrace{A \cdot n w kT}_{\boxed{}} = -\dot{V} n kT \\ &= -\frac{N}{V} \cdot \frac{dV}{dt} \end{aligned}$$

$$P = \frac{2}{3} \frac{E}{V} = n kT \quad \Rightarrow \quad DE = -\Delta V P \quad (VV)$$

for infinitesimal adiabatic change